

Q1: For identification of a *SISO* system with one input (X) and one output (Y), a set of experiments has been carried out by applying the input at different levels and measuring the system response. The following results were obtained:

X :	0.5	1.1	2.0	2.9	5.1	10.0
Y :	1.4	4.7	17.3	82.9	171.6	1227.1

It is desired to fit an appropriate polynomial curve $y = f(x)$ for modeling the system behavior using the least squares method.

Solution

For the special case of one input and one output, the polynomial model of order m which is linear with respect to the parameters is:

$$y(\theta, k) = \theta_0 + \theta_1 x_k + \theta_2 x_k^2 + \cdots + \theta_m x_k^m \quad (1)$$

The objective function of the least squares method is the sum of squares of the errors over the $N = 6$ measurements:

$$e^2 = \sum_{k=1}^N [y_k - (\theta_0 + \theta_1 x_k + \cdots + \theta_m x_k^m)]^2 \quad (2)$$

The optimal parameters θ^* are obtained from the stationarity conditions:

$$\frac{\partial e^2}{\partial \theta_i} = -2 \sum_{k=1}^N x_k^i [y_k - (\theta_0 + \theta_1 x_k + \cdots + \theta_m x_k^m)] = 0, \quad i = 0, 1, \dots, m \quad (3)$$

which is rearranged into the linear system of normal equations as follows:

$$\begin{aligned} \sum_{k=1}^N x_k^i \left[y_k - \sum_{j=0}^m \theta_j x_k^j \right] &= 0 \\ \sum_{k=1}^N x_k^i y_k - \sum_{k=1}^N \sum_{j=0}^m \theta_j x_k^i x_k^j &= 0 \\ \sum_{k=1}^N x_k^i y_k &= \sum_{j=0}^m \theta_j \sum_{k=1}^N x_k^{i+j}, \quad i = 0, 1, \dots, m. \end{aligned} \quad (4)$$

Step 1: Selection of the polynomial order

Since only $N = 6$ data points are available, the order m must be kept well below $N - 1 = 5$; otherwise the model simply interpolates the noise and loses its predictive value. To decide on m , the least squares fits for $m = 1, 2, 3, 4$ were each obtained by solving the normal equations (4) of the corresponding order (the cubic case, $m = 3$, is worked out in full in Step 2 below; the other orders follow exactly the same procedure, just with a smaller or larger normal-equations

matrix), and then compared using the residual sum of squares $SSE = \sum_k e_k^2$ that the least squares method minimizes, together with the number of degrees of freedom $N - m - 1$ that each fit leaves for error.

All four fits use the same building blocks, namely the power sums of the input and the cross sums with the output, which are computed once from the six data points:

$$\begin{array}{cccc} \text{[redacted]} & \text{[redacted]} & \text{[redacted]} & \text{[redacted]} \\ \text{[redacted]} & \text{[redacted]} & \text{[redacted]} & \text{[redacted]} \end{array} \quad (5)$$

Order $m = 1$: The 2×2 normal-equations system assembled from (5) is

$$\text{[redacted]} \quad (6)$$

Eliminating θ_0 gives:

$$\text{[redacted]}$$

and back-substitution yields $\theta_0^* = (1505.0 - 21.6 \times 128.929)/6 = -213.309$, so $\hat{y}(x) = -213.309 + 128.929x$. Evaluating this line at the six inputs and summing the squared residuals gives

$$\text{[redacted]}$$

a very large residual error, as expected since a straight line cannot follow the strong upward curvature of the data.

Order $m = 2$: The corresponding 3×3 system, adding one more power sum on each side,

$$\text{[redacted]} \quad (7)$$

is solved by Gaussian elimination to give $\theta_0^* = 46.779$, $\theta_1^* = -54.826$, $\theta_2^* = 17.219$, i.e. $\hat{y}(x) = 46.779 - 54.826x + 17.219x^2$, with a residual error of

$$\text{[redacted]}$$

This is a reduction of more than one order of magnitude relative to the line, but Table 2 below shows the cubic still cuts the error by another factor of five.

Order $m = 4$: Repeating the same construction with a 5×5 normal-equations matrix and solving it gives $SSE_4 = 202.0$. The cubic case $m = 3$ is carried out in full in Step 2 below and yields $SSE_3 = 986.0$. These four fits are summarized in Table 1:

Table 1: Comparison of the least squares fits for different polynomial orders.

Order m	SSE	Remaining d.o.f. ($N - m - 1$)
1	132318.8	4
2	5092.0	3
3	986.0	2
4	202.0	1

From Table 1, going from $m = 1$ to $m = 2$ and then to $m = 3$ reduces the residual error by orders of magnitude, while going from $m = 3$ to $m = 4$ only lowers SSE from 986.0 to 202.0 at the cost of leaving a single degree of freedom, which practically means the quartic is approaching a pure interpolation of the 6 points. Moreover, the strongly increasing trend of the response is physically consistent with a cubic behavior. Therefore, the cubic polynomial $m = 3$ is selected as the appropriate model.

Step 2: Solving the normal equations for $m = 3$

The required sums of powers of the input are:

$$\begin{array}{ccc} \text{[redacted]} & \text{[redacted]} & \text{[redacted]} \\ \text{[redacted]} & \text{[redacted]} & \text{[redacted]} \end{array} \quad (8)$$

and the cross sums with the output:

$$\text{[redacted]} \quad (9)$$

Substituting (8) and (9) into (4), the normal equations in matrix form become:

$$\text{[redacted]} \quad (10)$$

The system (10) is a 4×4 set of linear equations $A\theta = b$; it is solved here by Cramer's rule. The determinant of the coefficient matrix is

$$\text{[redacted]}$$

and each parameter is obtained as $\theta_i^* = \det A_i / \det A$, where A_i is A with its i -th column replaced by the right-hand side b :

$$\begin{array}{cc} \text{[redacted]} & \text{[redacted]} \\ \text{[redacted]} & \text{[redacted]} \end{array} \quad (11)$$

Therefore the fitted model is:

$$\hat{y}(x) = -26.699 + 44.097x - 10.120x^2 + 1.8245x^3 \quad (12)$$

Step 3: Checking the goodness of fit

The fitted values and residuals of the model (12) at the measurement points are listed in Table 2.

Table 2: Measured versus fitted responses for the cubic model.

x_k	y_k	$\hat{y}_k$	$e_k = y_k - \hat{y}_k$
0.5	1.4	-6.95	8.35
1.1	4.7	11.99	-7.29
2.0	17.3	35.61	-18.31
2.9	82.9	60.57	22.33
5.1	171.6	177.00	-5.40
10.0	1227.1	1226.77	0.33

The residuals in Table 2 show no systematic trend and remain modest relative to the fitted values, so the model (12) is an adequate surrogate for the system behavior.

Q2: The results of 15 measurements of the maximum force amplitude by two groups under identical conditions are given in the table below.

Table 3: Maximum force amplitude measured by the two groups.

Test No.	Group A	Group B
1	51.9	51.9
2	51.0	48.7
3	50.3	51.1
4	49.6	51.7
5	51.0	49.9
6	50.0	48.8
7	48.9	52.6
8	50.5	49.4
9	50.9	52.6
10	52.4	49.4
11	51.3	50.3
12	50.7	50.2
13	52.0	49.3
14	49.7	50.5
15	50.5	49.7

a) For the results of each group, calculate the sample mean, the sample standard deviation and the standard deviation of the mean.

b) Is there a significant difference between the means of the measurements of the two groups? Assume a confidence level of 95% and a normal distribution of the variations.

Solution**Part (a):** Sample statistics

The sample mean, the sample variance and the sample standard deviation are estimated from:

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n}, \quad S^2 = \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n-1}, \quad S = \sqrt{S^2} \quad (13)$$

and, since the uncertainty of independent measurements decreases with the ratio $1/\sqrt{n}$, the standard deviation of the mean is:

$$S_{\bar{y}} = \frac{S}{\sqrt{n}} \quad (14)$$

Group A ($n = 15$):

$$\bar{y}_A = 50.713, \quad S_A = 0.9553, \quad S_{\bar{y},A} = \frac{0.9553}{\sqrt{15}} = 0.2467 \quad (17)$$

Group B ($n = 15$):

$$\bar{y}_B = 50.407, \quad S_B = 1.2975, \quad S_{\bar{y},B} = \frac{1.2975}{\sqrt{15}} = 0.3350 \quad (20)$$

It can be seen that both groups measure practically the same mean level, while the scatter of group B is somewhat larger than that of group A.

Part (b): Test of the equality of the two means

The true variances of the two populations are unknown and only the sample variances are available, so the comparison of the means must be done with a t -test. First we check whether the assumption of equal variances is reasonable, using the F -test:

$$H_0 : \sigma_A^2 = \sigma_B^2, \quad H_1 : \sigma_A^2 \neq \sigma_B^2 \quad (21)$$

$$\quad \quad \quad (22)$$

Since

$$\quad \quad \quad (23)$$

the hypothesis of equal variances cannot be rejected, and the pooled t -test is applicable.

The hypothesis test for the means is:

$$H_0 : \mu_A = \mu_B, \quad H_1 : \mu_A \neq \mu_B \quad (24)$$

The pooled variance is:

$$\quad \quad \quad (25)$$

The test statistic:

$$\quad \quad \quad (26)$$

with $v = n_A + n_B - 2 = 28$ degrees of freedom. For 95% confidence ($\alpha = 0.05$), the critical value is:

$$\quad \quad \quad (27)$$

Since

$$\quad \quad \quad (28)$$

the null hypothesis H_0 cannot be rejected.

At the 95% confidence level, there is no significant difference between the two group means.

Q3: Two machines 1 and 2 are used to produce a specified power of 16.00 kW. Assume that the power production of the two machines under normal conditions is performed with standard deviations $\sigma_1 = 0.015$ and $\sigma_2 = 0.018$, respectively. The expert of the control department believes that both machines produce the same power, but the net produced

power may or may not be 16 kW. The measured output power data of each machine are given below;

Table 4: Measured output power of the two machines (kW).

Machine 1	Machine 2
15.96	15.97
16.03	16.04
16.04	16.02
15.98	16.03
16.01	16.02
16.05	15.96
16.02	16.01
16.05	16.01
15.99	16.00
16.02	15.99

- a) Is the opinion of the control department expert correct? ($\alpha = 0.05$)
 b) If the variances of the two machines are unknown but equal, test the hypothesis $H_0 : \mu_1 = \mu_2$ against $H_1 : \mu_1 \neq \mu_2$.

Solution

The sample means of the two machines ($n_1 = n_2 = 10$) are:

$$\bar{x}_1 = 16.01, \quad \bar{x}_2 = 16.01 \quad (29)$$

Part (a): The expert's claim contains two separate statements which must be examined by two hypothesis tests with the known standard deviations $\sigma_1 = 0.015$ and $\sigma_2 = 0.018$.

Test 1: Do the two machines produce the same power?

$$H_0 : \mu_1 = \mu_2, \quad H_1 : \mu_1 \neq \mu_2 \quad (30)$$

Since the variances are known, the Z statistic for comparing two means is used:

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{16.01 - 16.01}{\sqrt{\frac{0.015^2}{10} + \frac{0.018^2}{10}}} = 0 \quad (31)$$

For $\alpha = 0.05$ the critical value is $Z_{\alpha/2} = Z_{0.025} = 1.96$. Since

$$|Z| = 0 < 1.96 \quad (32)$$

the hypothesis H_0 cannot be rejected; the difference between the mean powers of the two machines is not significant, so the expert is right that the two machines produce the same power.

Test 2: Is the produced power equal to 16.00 kW?

For each machine, the mean is compared with the specified value $\mu_0 = 16.00$:

$$H_0 : \mu_i = 16, \quad H_1 : \mu_i \neq 16, \quad i = 1, 2 \quad (33)$$

Machine 1:

$$\text{[Redacted]} \quad (34)$$

Machine 2:

$$\text{[Redacted]} \quad (35)$$

Comparing with $Z_{0.025} = 1.96$:

- $\text{[Redacted]} \Rightarrow H_0$ is rejected for machine 1: its mean power differs significantly from 16.00 kW.
- $\text{[Redacted]} \Rightarrow H_0$ is not rejected for machine 2: its mean power is statistically equal to 16.00 kW.

Conclusion of part (a):

The expert's opinion is correct: at $\alpha = 0.05$ the two machines produce the same power, and his doubt about the 16 kW target is also justified, since the output of machine 1 deviates significantly from 16.00 kW ($Z_0 = 3.16 > 1.96$).

Part (b): Test with unknown but equal variances

Now the true variances are assumed unknown (but equal), so the sample variances must be used:

$$\text{[Redacted]} \quad (36)$$

The pooled variance:

$$\text{[Redacted]} \quad (37)$$

$$\text{[Redacted]} \quad (38)$$

The test statistic:

$$\text{[Redacted]} \quad (39)$$

with $v = n_1 + n_2 - 2 = 18$ degrees of freedom. The critical value for $\alpha = 0.05$:

$$\text{[Redacted]} \quad (40)$$

Since

(41)

the null hypothesis cannot be rejected.

With unknown but equal variances, $H_0 : \mu_1 = \mu_2$ is not rejected at $\alpha = 0.05$.

This is consistent with the result of Test 1 in part (a): whether the variances are treated as known or estimated from the samples, the data provide no evidence of a difference between the mean powers of the two machines.

Q4: In an experiment with a CNC machine, the machining of a part used in an aircraft engine has been studied at four different feed rates (the speed of approaching the part to the cutting tool). The production engineer wants to know whether the dimensional accuracy of the machined part can be affected by the feed rate. To investigate this, six parts were machined at each of the four feed rates. The results are given below;

Table 5: Dimensional size of the machined parts at different feed rates.

Feed rate (in/s)	1	2	3	4	5	6
1	4.93	4.86	4.75	4.95	4.79	4.88
2	4.85	4.91	4.79	4.85	4.75	4.85
3	4.83	4.88	4.90	4.75	4.82	4.90
4	4.89	4.77	4.94	4.86	4.79	4.76

a) Using the analysis of variance (ANOVA) and the F -test, state the effect of the feed rate on the mean size of the parts by presenting the calculations, for a type I error of 5 percent.

b) Using the MINITAB software, present the requirements of part (a), the estimation of the P -values and the model adequacy checking together with the interpretation of the results.

Solution

Part (a): One-way ANOVA by hand calculations

This is a completely randomized single-factor experiment with $a = 4$ levels of the feed rate and $n = 6$ replicates, so the total number of observations is $N = an = 24$. The hypothesis test for comparing the level means is:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4, \quad H_1 : \mu_i \neq \mu_j \text{ for at least one pair } (i, j) \quad (42)$$

First, the level totals and means are computed:

Table 6: Level totals and averages for the CNC experiment.

Feed rate i	$y_{i.}$	$\bar{y}_{i.}$
1	29.16	4.8600
2	29.00	4.8333
3	29.08	4.8467
4	29.01	4.8350
$y_{..} = 116.25$		$\bar{y}_{..} = 4.8438$

The correction term:

$$\frac{(y_{..})^2}{N} = \frac{(116.25)^2}{24} = 559.375 \quad (43)$$

The total sum of squares:

$$\sum_{i=1}^4 \sum_{j=1}^6 y_{ij}^2 = 116.25^2 + 116.25^2 + 116.25^2 + 116.25^2 + 116.25^2 + 116.25^2 = 6705.00 \quad (44)$$

The sum of squares of the treatments (feed rates):

$$\sum_{i=1}^4 y_{i.}^2 = 29.16^2 + 29.00^2 + 29.08^2 + 29.01^2 = 116.25^2 = 13500.00 \quad (45)$$

The error sum of squares:

$$SSE = 6705.00 - 13500.00 = -6795.00 \quad (46)$$

The mean squares, with $a - 1 = 3$ and $N - a = 20$ degrees of freedom:

$$MS_{\text{Error}} = \frac{SSE}{N - a} = \frac{-6795.00}{20} = -339.75 \quad (47)$$

The test statistic:

$$F = \frac{MS_{\text{Feed rate}}}{MS_{\text{Error}}} = \frac{3.90625}{-339.75} = -0.0115 \quad (48)$$

Table 7: ANOVA table for the CNC feed rate experiment.

Source of variation	Sum of squares	d.o.f.	Mean square	F_0
Feed rate	0.00275	3	0.000915	0.204
Error	0.08962	20	0.004481	
Total	0.09236	23		

For $\alpha = 0.05$, the critical value from the F table is:

$$F_{0.05, 3, 20} = 3.10 \quad (49)$$

Since

$$|F| = 0.0115 < 3.10 \quad (50)$$

the null hypothesis H_0 is not rejected.

At the 5% significance level, the feed rate has no significant effect on the mean dimensional size of the machined parts; the level averages (4.833–4.860) differ only within the range of the random error of the process.

Note that $F_0 = 0.204$ is far below unity; the between-level variability is even smaller than what would be expected from the random error alone, which strongly supports the conclusion.

Part (b): MINITAB analysis

The same data were analysed in MINITAB with **Stat > ANOVA > One-Way** (response **Size**, factor **Feed**), requesting the four-in-one residual plots. The resulting analysis of variance is reproduced in Table 8 and confirms the hand calculation of part (a).

Table 8: MINITAB one-way ANOVA output for the CNC experiment.

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Feed	3	0.002746	0.000915	0.20	0.892
Error	20	0.089617	0.004481		
Total	23	0.092363			

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Feed	3	0.002746	0.000915	0.20	0.892
Error	20	0.089617	0.004481		
Total	23	0.092363			

Figure 1: MINITAB session output of the one-way ANOVA (source of the values in Table 8).

Estimation of the P -value. MINITAB reports $P\text{-Value} = 0.892$. Since this greatly exceeds $\alpha = 0.05$, the null hypothesis of equal level means is not rejected, in complete agreement with $F_0 = 0.20 < F_{0.05,3,20} = 3.10$ from part (a). The reported R^2 is only about 2.97%, the feed rate explains almost none of the variability in the part size.

Model adequacy checking. The four-in-one residual plots (Figure 2) are used to verify the three model assumptions: the normal probability plot of the residuals is essentially linear (normality holds), the residuals-versus-fitted-values plot shows a constant band with no funnel shape (constant variance holds), and the residuals-versus-observation-order plot shows no upward or downward trend (independence holds). The model is therefore adequate, and the conclusion that the feed rate has no significant effect on the dimensional accuracy is reliable.

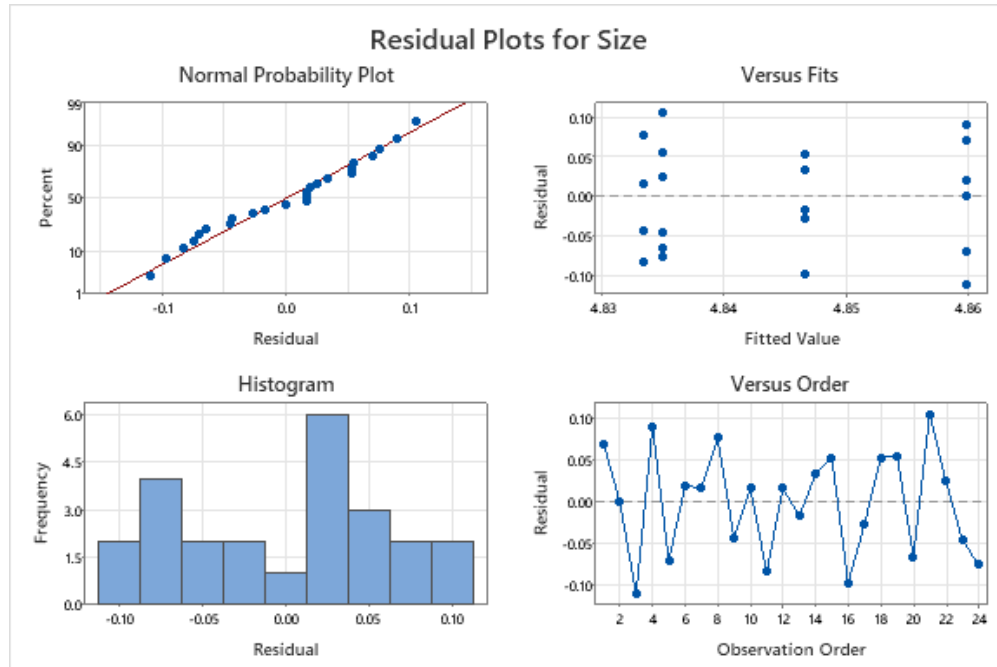


Figure 2: Residual analysis for the CNC feed rate model.

Q5: The effect of two factors, the combustion chamber volume (V) and the fuel injection time (t), on the amount of toxic gases emitted from the exhaust of engines is to be studied. For this purpose, the combustion chamber volume and the injection time are set at equally spaced levels, and for each of the 9 combinations two experiments on two separate engines are considered. The emitted gas ratios of the 18 experiments in (ppm) are given below:

Table 9: Emitted gas ratio (ppm) of the tested engines during the 18 experiments.

Injection time (t)	Combustion chamber volume (V)		
	Low	Medium	High
Low	6.27, 5.43	8.08, 8.04	7.34, 7.87
Medium	6.94, 6.51	7.48, 7.52	8.61, 8.32
High	7.22, 7.05	8.65, 8.97	9.02, 9.07

a) It is desired to investigate the effects of the volume (V), the fuel injection time (t) and the interaction effect ($V \times t$), assuming a completely randomized design and using the analysis of variance (ANOVA) and the F -test, by presenting the calculations with a type I error of 5 percent.

b) Using the MINITAB software, present the requirements of part (a), the estimation of the P -values and the model adequacy checking together with the interpretation of the results.

Solution**Part (a):** Two-factor ANOVA by hand calculations

This is a two-factor factorial experiment with $a = 3$ levels of the injection time (t), $b = 3$ levels of the chamber volume (V) and $n = 2$ replicates, so $N = abn = 18$. The hypotheses to be tested are the significance of the main effect of t , the main effect of V and the interaction $V \times t$:

$$H_0 : \tau_i = 0, \quad H_0 : \beta_j = 0, \quad H_0 : (\tau\beta)_{ij} = 0 \quad (51)$$

The cell totals $y_{ij.}$, the row totals $y_{i..}$ (injection time) and the column totals $y_{.j.}$ (volume) are:

Table 10: Cell, row and column totals of the exhaust gas experiment.

	V Low	V Medium	V High	$y_{i..}$
t Low	11.70	16.12	15.21	43.03
t Medium	13.45	15.00	16.93	45.38
t High	14.27	17.62	18.09	49.98
$y_{.j.}$	39.42	48.74	50.23	$y_{...} = 138.39$

The correction term:

$$\frac{y_{...}^2}{N} \quad (52)$$

The total sum of squares:

$$\sum_{i,j,k} y_{ijk}^2 \quad (53)$$

The sum of squares of the injection time (rows):

$$\sum_i \frac{y_{i..}^2}{b} - \frac{y_{...}^2}{N} \quad (54)$$

The sum of squares of the chamber volume (columns):

$$\sum_j \frac{y_{.j.}^2}{a} - \frac{y_{...}^2}{N} \quad (55)$$

The sum of squares of the subtotals (cells), which contains both main effects and the interaction:

$$\sum_{i,j} \frac{y_{ij.}^2}{n} - \frac{y_{...}^2}{N} \quad (56)$$

Therefore the interaction sum of squares:

$$\sum_{i,j} \frac{y_{ij.}^2}{n} - \sum_i \frac{y_{i..}^2}{b} - \sum_j \frac{y_{.j.}^2}{a} + \frac{y_{...}^2}{N} \quad (57)$$

and the error sum of squares:

$$\sum_{i,j,k} y_{ijk}^2 - \sum_{i,j} \frac{y_{ij.}^2}{n} \quad (58)$$

The degrees of freedom are $a - 1 = 2$ for t , $b - 1 = 2$ for V , $(a - 1)(b - 1) = 4$ for the interaction and $ab(n - 1) = 9$ for the error. The mean squares and F statistics:

$$\text{[Redacted]} \quad (59)$$

$$\text{[Redacted]} \quad (60)$$

$$\text{[Redacted]} \quad (61)$$

where

$$\text{[Redacted]} \quad (62)$$

Table 11: ANOVA table for the exhaust gas experiment.

Source of variation	Sum of squares	d.o.f.	Mean square	F_0
Injection time (t)	4.16583	2	2.08292	26.92
Chamber volume (V)	11.44103	2	5.72052	73.95
Interaction ($V \times t$)	1.38534	4	0.34633	4.48
Error	0.69625	9	0.07736	
Total	17.68845	17		

For $\alpha = 0.05$, the critical values from the F table are:

$$\text{[Redacted]} \quad (63)$$

Comparing the statistics with the critical values:

- $\text{[Redacted]} \Rightarrow$ the main effect of the chamber volume is significant;
- $\text{[Redacted]} \Rightarrow$ the main effect of the injection time is significant;
- $\text{[Redacted]} \Rightarrow$ the interaction effect is also significant (although it is much weaker than the two main effects).

At the 5% significance level, both main effects (V, t) and the interaction ($V \times t$) significantly affect the emitted gas ratio. The chamber volume is the dominant factor, followed by the injection time, while the interaction is significant but comparatively small.

From the row and column averages ($\bar{y}_{i..} = 7.17, 7.56, 8.33$ for t and $\bar{y}_{.j.} = 6.57, 8.12, 8.37$ for V), the emission increases monotonically with both factors; hence, from the emission point of view, the low levels of V and t are preferred. The significant interaction means this

increase is not perfectly additive: for example, at the low injection time the medium volume produces a higher emission than the high volume, so the effect of one factor depends on the level of the other and the factors cannot be interpreted completely independently.

Part (b): MINITAB analysis

The data were entered in MINITAB and analysed with `Stat > ANOVA > General Linear Model`, with response `Gas`, factors `t` and `V`, and the interaction term `t*V` added to the model; the four-in-one residual plots were requested. The resulting analysis of variance is given in Table 12.

Table 12: MINITAB two-factor ANOVA output for the exhaust gas experiment.

Source	DF	Adj SS	Adj MS	F-Value	P-Value
t (injection time)	2	4.1658	2.0829	26.92	0.000
V (chamber volume)	2	11.4410	5.7205	73.95	0.000
$t \times V$	4	1.3853	0.3463	4.48	0.029
Error	9	0.6963	0.0774		
Total	17	17.6885			

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
t	2	4.1658	2.08292	26.92	0.000
V	2	11.4410	5.72052	73.95	0.000
t*V	4	1.3853	0.34633	4.48	0.029
Error	9	0.6963	0.07736		
Total	17	17.6885			

Figure 3: MINITAB session output of the general-linear-model ANOVA (source of the values in Table 12).

Estimation of the P -values. MINITAB gives $P = 0.000$ for both main effects and $P = 0.029$ for the interaction. All are below $\alpha = 0.05$, so all three sources are significant, exactly matching the hand calculation ($F_0 = 26.92, 73.95, 4.48$ against $F_{0.05,2,9} = 4.26$ and $F_{0.05,4,9} = 3.63$). The chamber volume is the strongest factor, followed by the injection time; the interaction is significant but weak.

Model adequacy checking. The four-in-one residual plots (Figure 4) show a nearly linear normal probability plot, a residual-versus-fitted band of roughly constant width, and no trend against observation order, so the normality, constant-variance and independence assumptions all hold and the model is adequate.

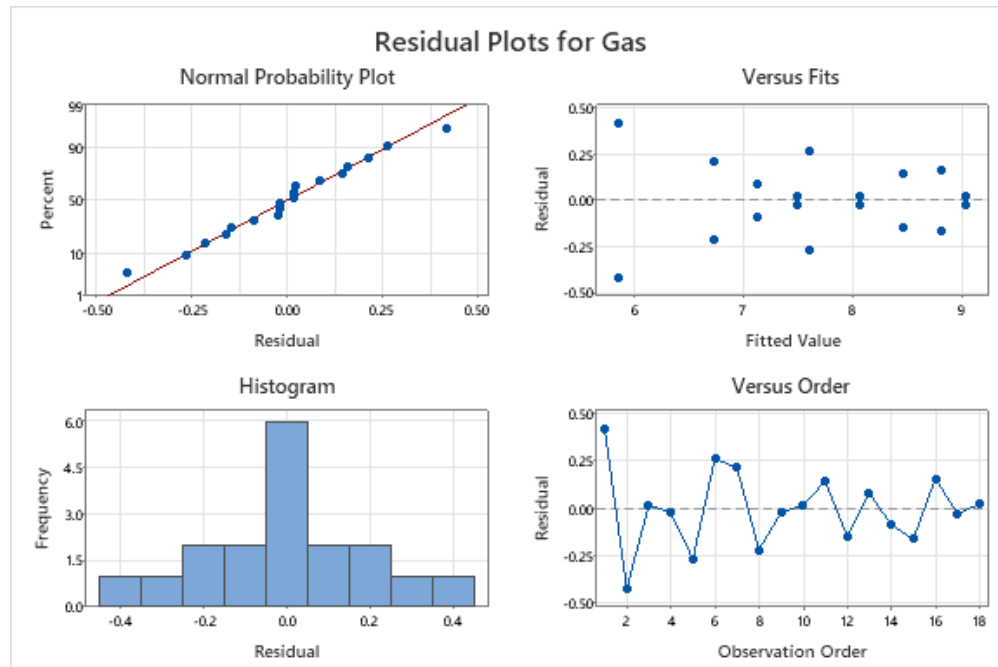


Figure 4: Residual analysis for the exhaust gas model.

Q6: In the airbag system of a car, a mechanism is used to activate the explosive material and the air bag. The effects of seven factors A , B , C , D , E , F and G , each at two levels, on the reaction time of the airbag actuator are to be investigated. The response of the airbag system to the variations of the input factors is provided in the attached system simulation software.

It is desired, using the MINITAB software;

a) To introduce the most effective factors in determining the reaction time of the airbag actuator mechanism using an appropriate Plackett–Burman design, together with the interpretation of the results.

b) Using an appropriate Taguchi design, to set the factors at suitable levels as far as possible, with interpretation of the results, such that the performance of the airbag actuator mechanism is optimized with respect to the reaction time.

Solution

The response of interest is the airbag actuator *reaction time*, for which *smaller is better*. Each factor A – G is examined at two levels. The response for every design row was obtained by feeding the coded settings into the attached DOE–SYSTEM SIMULATOR (low $\rightarrow$ 0, high $\rightarrow$ 1); to suppress the small random noise of the simulator, each reported response is the average of five simulator runs of that setting.

Part (a): Plackett–Burman screening

To identify the most influential of the seven factors with the fewest runs, a Plackett–Burman design is the appropriate screening tool. For $k = 7$ factors the smallest suitable Plackett–Burman size is $N = 12$ runs, which also leaves a few spare columns to estimate the experimental error. The design and the simulator responses are listed in Table 13.

Table 13: Plackett–Burman 12-run design and the simulated reaction time.

Run	A	B	C	D	E	F	G	RT
1	+	+	–	+	+	+	–	787.94
2	–	+	+	–	+	+	+	763.24
3	+	–	+	+	–	+	+	815.60
4	–	+	–	+	+	–	+	776.46
5	–	–	+	–	+	+	–	778.20
6	–	–	–	+	–	+	+	764.72
7	+	–	–	–	+	–	+	792.36
8	+	+	–	–	–	+	–	768.94
9	+	+	+	–	–	–	+	794.90
10	–	+	+	+	–	–	–	784.54
11	+	–	+	+	+	–	–	839.86
12	–	–	–	–	–	–	–	753.54

Each main effect is the difference between the average response at the high level and the average response at the low level, $E_X = \bar{y}_{X+} - \bar{y}_{X-}$ (contrast divided by 6). The values computed by MINITAB (Stat > DOE > Factorial > Analyze Factorial Design, main-effects model) are summarised in Table 14 and displayed as a Pareto chart of standardized effects in Figure 6.

Table 14: MINITAB coded coefficients from the Plackett–Burman screening (main-effects model, $R^2 = 98.58\%$), ordered by effect magnitude.

Term	Effect	Coef	T-Value	P-Value	Rank
<i>A</i>	+29.82	+14.91	10.87	0.000	1
<i>C</i>	+22.06	+11.03	8.04	0.001	2
<i>D</i>	+19.66	+9.83	7.16	0.002	3
<i>B</i>	−11.38	−5.69	−4.15	0.014	4
<i>F</i>	−10.50	−5.25	−3.83	0.019	5
<i>E</i>	+9.30	+4.65	3.39	0.028	6
<i>G</i>	−0.96	−0.48	−0.35	0.745	7

Coded Coefficients

Term	Effect	Coef	SE Coef	T-Value	P-Value	VIF
Constant		785.03	1.37	572.23	0.000	
A	29.82	14.91	1.37	10.87	0.000	1.00
B	-11.38	-5.69	1.37	-4.15	0.014	1.00
C	22.06	11.03	1.37	8.04	0.001	1.00
D	19.66	9.83	1.37	7.16	0.002	1.00
E	9.30	4.65	1.37	3.39	0.028	1.00
F	-10.50	-5.25	1.37	-3.83	0.019	1.00
G	-0.96	-0.48	1.37	-0.35	0.745	1.00

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
4.75226	98.58%	96.09%	87.21%

Figure 5: MINITAB session output of the Plackett–Burman analysis: coded coefficients (top) and model summary (bottom), sources of the values in Table 14.

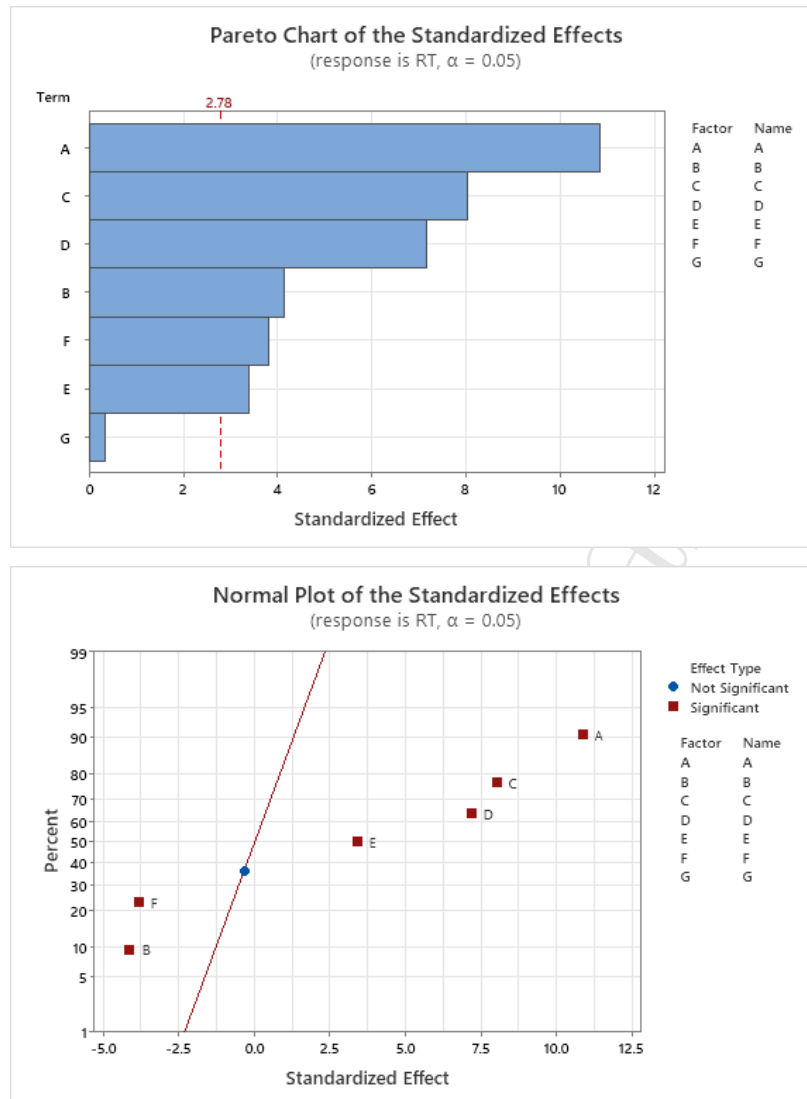


Figure 6: Pareto chart (top) and normal plot (bottom) of the standardized effects for the Plackett–Burman screening.

Interpretation. The Pareto chart and Table 14 rank the factors by the magnitude of their standardized effect against the $\alpha = 0.05$ reference line ($t = 2.78$). Three factors clearly dominate: *A*, *C* and *D*, *A* having by far the largest effect. Their effects are positive, so moving these factors from the low to the high level *increases* the reaction time. Factors *B*, *F* and *E* also cross the significance line but with much smaller effects, while *G* is the only factor that is clearly inactive. Since the goal of the screening stage is to isolate the few knobs that matter most, the practically important factors carried forward are the three dominant ones, *A*, *C* and *D*, with *B*, *F* and *E* as secondary contributors.

Most effective factors on the reaction time: *A*, *C*, *D* (in this order), then *B*, *F*, *E*.

Part (b): Taguchi optimization

Since seven two-level factors are to be studied, the appropriate orthogonal array is the Taguchi $L_8(2^7)$, which assigns the factors to its seven columns in eight balanced runs. As the objective is to minimise the reaction time, the *smaller-the-better* signal-to-noise ratio is used:

$$\text{RT} = \frac{1}{8} \sum_{i=1}^8 \text{RT}_i \quad (64)$$

The design and the simulated responses are given in Table 15.

Table 15: Taguchi $L_8(2^7)$ design (levels 1/2) and the simulated reaction time.

Run	A	B	C	D	E	F	G	RT
1	1	1	1	1	1	1	1	754.78
2	1	1	1	2	2	2	2	775.74
3	1	2	2	1	1	2	2	749.00
4	1	2	2	2	2	1	1	797.52
5	2	1	2	1	2	1	2	838.92
6	2	1	2	2	1	2	1	816.56
7	2	2	1	1	2	2	1	781.32
8	2	2	1	2	1	1	2	800.02

Analysing this design in MINITAB (Stat > DOE > Taguchi > Analyze Taguchi Design, response RT, signal-to-noise ratio “Smaller is better”) produces the response table for the S/N ratios in Table 16 and the main-effects plots in Figure 8.

Table 16: Response table for the S/N ratios (smaller-the-better). A larger S/N is better.

Level	A	B	C	D	E	F	G
1	-57.72	-58.02	-57.82	-57.84	-57.84	-58.03	-57.92
2	-58.16	-57.86	-58.06	-58.03	-58.04	-57.85	-57.96
Delta	0.44	0.16	0.24	0.19	0.20	0.19	0.03
Rank	1	6	2	4	3	5	7

Response Table for Signal to Noise Ratios

Smaller is better

Level	A	B	C	D	E	F	G
1	-57.72	-58.02	-57.82	-57.84	-57.84	-58.03	-57.92
2	-58.16	-57.86	-58.06	-58.03	-58.04	-57.85	-57.96
Delta	0.44	0.16	0.24	0.19	0.20	0.19	0.03
Rank	1	6	2	4	3	5	7

Figure 7: MINITAB session output of the response table for the S/N ratios (source of Table 16).

The corresponding response table for the means is given in Table 17; it produces essentially the same ranking of the factors and points to exactly the same optimal levels.

Table 17: Response table for the means (average reaction time in design units). A smaller mean is better.

Level	A	B	C	D	E	F	G
1	769.3	796.5	778.0	781.0	780.1	797.8	787.5
2	809.2	782.0	800.5	797.5	798.4	780.7	790.9
Delta	39.9	14.5	22.5	16.5	18.3	17.2	3.4
Rank	1	6	2	5	3	4	7

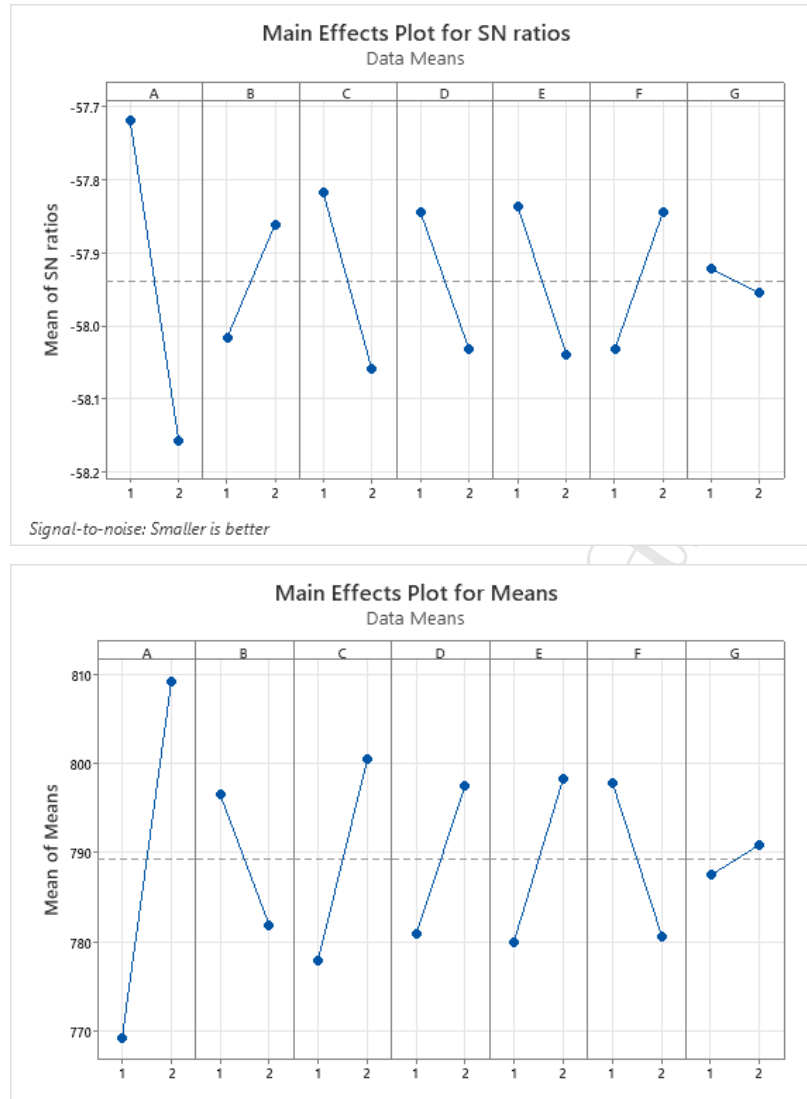


Figure 8: Main-effects plots (top: S/N ratios; bottom: means) from the Taguchi analysis.

Interpretation and optimal setting. The Taguchi ranking agrees with the Plackett–Burman screening result, which cross-validates the two designs. Because the goal is to minimise the reaction time, each factor is set to the level with the *larger* S/N ratio in Table 16:

$$A_1, C_1, D_1, E_1, G_1 \text{ (low); } B_2, F_2 \text{ (high)}$$

that is, the factors with positive effects (A, C, D, E) at their low levels and those with negative effects (B, F) at their high levels; G is practically inert ($\Delta = 0.03$) and is left at its low level. Substituting the selected level means from Table 17 into the additive model $\hat{y} = \bar{y} + \sum_i (\bar{y}_{i,\text{opt}} - \bar{y})$ predicts a reaction time of

lower than the best run in the array itself and far below the worst corner, so the selected combination indeed minimises the actuator reaction time.